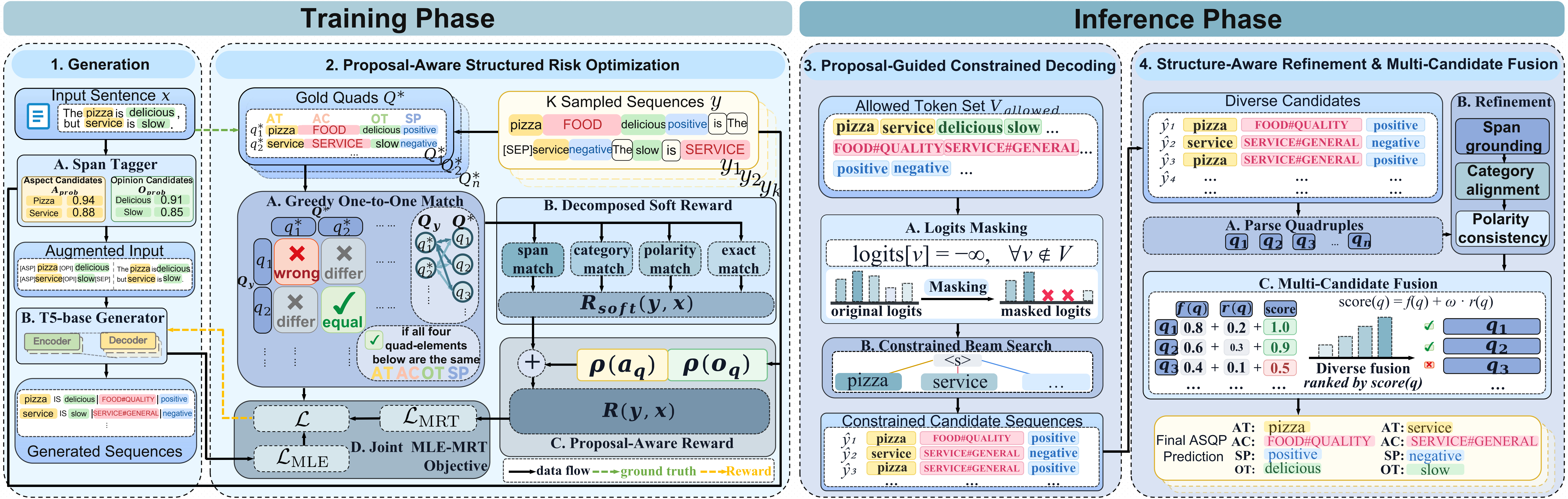
Training Phase
Inference Phase
1. Generation
Input Sentence x
The pizza is delicious, but service is slow.
A. Span Tagger
Aspect Candidates A_{prob}
Pizza 0.94
Service 0.88
Opinion Candidates O_{prob}
Delicious 0.91
Slow 0.85
Augmented Input
[ASP] pizza [OPI] delicious [ASP] service [OPI] slow [SEP] The pizza is delicious but service is slow
B. T5-base Generator
Encoder
Decoder
pizza IS delicious | FOOD#QUALITY | positive
service IS slow | SERVICE#GENERAL | negative
Generated Sequences
2. Proposal-Aware Structured Risk Optimization
Gold Quads Q^*
AT AC OT SP
q_1^* pizza FOOD delicious positive
q_2^* service SERVICE slow negative
Q_1^* Q_2^* Q_n^*
K Sampled Sequences y
pizza FOOD delicious positive is The
[SEP] service negative The slow is SERVICE
y_1 y_2 y_k
A. Greedy One-to-One Match
Q^* q_1^* q_2^*
Q_y q_1 q_2
wrong
differ
differ
equal
Q_y Q^*
if all four quad-elements below are the same
AT AC OT SP
B. Decomposed Soft Reward
span match
category match
polarity match
exact match
$R_{soft}(y, x)$
$\rho(a_q)$
$\rho(o_q)$
$R(y, x)$
C. Proposal-Aware Reward
$\mathcal{L}$
$\mathcal{L}_{MRT}$
$\mathcal{L}_{MLE}$
D. Joint MLE-MRT Objective
data flow
ground truth
Reward
3. Proposal-Guided Constrained Decoding
Allowed Token Set $V_{allowed}$
pizza service delicious slow ...
FOOD#QUALITY SERVICE#GENERAL ...
positive negative ...
A. Logits Masking
$\text{logits}[v] = -\infty, \quad \forall v \notin V$
original logits
Masking
masked logits
B. Constrained Beam Search
<s>
pizza
service
...
Constrained Candidate Sequences
$\hat{y}_1$ pizza FOOD#QUALITY positive
$\hat{y}_2$ service SERVICE#GENERAL negative
$\hat{y}_3$ pizza SERVICE#GENERAL positive
4. Structure-Aware Refinement & Multi-Candidate Fusion
Diverse Candidates
$\hat{y}_1$ pizza FOOD#QUALITY positive
$\hat{y}_2$ service SERVICE#GENERAL negative
$\hat{y}_3$ pizza SERVICE#GENERAL positive
$\hat{y}_4$ ...
A. Parse Quadruples
q_1 q_2 q_3 ... q_n
B. Refinement
Span grounding
Category alignment
Polarity consistency
C. Multi-Candidate Fusion
$score(q) = f(q) + \omega \cdot r(q)$
$f(q)$ $r(q)$ score
q_1 0.8 + 0.2 + 1.0 ✓
q_2 0.6 + 0.3 + 0.9 ✓
q_3 0.4 + 0.1 + 0.5 ☒
Diverse fusion ranked by score(q)
q_1 q_2 q_3
Final ASQP Prediction
AT: pizza AC: FOOD#QUALITY SP: positive OT: delicious
AT: service AC: SERVICE#GENERAL SP: negative OT: slow